

Thermodynamic origin of the vacuum–matter coupling from de Sitter two-fluid hydrodynamics

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Abstract

Volovik’s two-fluid thermodynamics of the de Sitter state treats the gravitational coupling $K = 1/16\pi G$ and the scalar curvature R as conjugate thermodynamic variables, with the Gibbs–Duhem relation $Ts = KR$ governing the gravitational (normal) component of the cosmological fluid. We show that self-consistent inclusion of a local matter perturbation ρ_m into this framework, under the condition that the de Sitter temperature $T = H/\pi$ is set by the cosmological background and not by local matter content, requires the vacuum energy density to shift as $\delta\varepsilon_{\text{vac}} = -\alpha_{\text{th}}\rho_m$ with $\alpha_{\text{th}} = 1/4$, and simultaneously requires the gravitational coupling K to vary locally. When both T and K are allowed to vary, the Gibbs–Duhem relation admits a one-parameter family of solutions; the physical condition $\delta T = 0$ for local perturbations selects $\alpha = 1/4$ uniquely. The QCD chiral sector, through the pion–nucleon sigma term, contributes $\alpha_\chi = \sigma_{\pi N}/m_N \approx 0.05$, which is 20% of the thermodynamic value. This identifies the chiral fraction of the vacuum response and suggests that the remaining $\sim 80\%$ may reside in the gluonic sector.

1 Introduction

The gravitating vacuum model [1] proposes that the vacuum energy density responds to the local matter density:

$$\rho_{\text{vac}}(\rho_m) = \Lambda_0 - \alpha \rho_m, \quad (1)$$

where Λ_0 is the bare cosmological constant and α is a dimensionless coupling. The coupling has been derived from QCD: the in-medium shift of the chiral condensate, quantified by the pion–nucleon sigma term $\sigma_{\pi N} \approx 50$ MeV [2], gives

$$\alpha_\chi = \frac{\sigma_{\pi N}}{m_N} \approx 0.05, \quad (2)$$

where m_N is the nucleon mass. This microscopic derivation provides a specific numerical value, but lacks a thermodynamic foundation: it does not explain *why* the vacuum should respond to matter, only *how much* the chiral sector responds.

Independently, Volovik has developed a thermodynamic description of the de Sitter state [3, 4, 5] in which the gravitational coupling $K = 1/16\pi G$ and the scalar curvature R form a pair of conjugate thermodynamic variables. The de Sitter vacuum is treated as a two-component fluid: a “superfluid” dark energy component with equation of state $w = -1$, and a thermal gravitational component with $w = +1$, whose Gibbs–Duhem relation is

$$Ts = KR, \quad (3)$$

where $T = H/\pi$ is the local de Sitter temperature and s is the entropy density of the gravitational component.

In this paper we show that self-consistent inclusion of matter into Volovik’s thermodynamic framework, combined with the physical condition that local matter perturbations do not modify the cosmological expansion rate, determines the vacuum–matter coupling: $\alpha_{\text{th}} = 1/4$ at linear order. The ratio $\alpha_\chi/\alpha_{\text{th}} = 0.2$ then identifies the chiral fraction of the vacuum response.

2 De Sitter thermodynamics: review

We work in units $\hbar = c = 1$ and keep G explicit unless stated otherwise. The de Sitter state with Hubble parameter H has the following thermodynamic quantities [3, 5]:

$$\varepsilon_{\text{vac}} = \frac{3H^2}{8\pi G}, \quad (4)$$

$$T = \frac{H}{\pi}, \quad (5)$$

$$s = \frac{3H}{4G} = \frac{3\pi T}{4G}, \quad (6)$$

$$K = \frac{1}{16\pi G}, \quad R = 12H^2. \quad (7)$$

The vacuum energy density can be written as

$$\varepsilon_{\text{vac}} = 6\pi^2 K T^2. \quad (8)$$

The entropy density is obtained from the free energy $F = -\varepsilon_{\text{vac}}$ (valid for $\varepsilon \propto T^2$):

$$s = -\frac{\partial F}{\partial T} = \frac{2\varepsilon_{\text{vac}}}{T} = 12\pi^2 K T. \quad (9)$$

The Gibbs–Duhem relation for the gravitational component is readily verified:

$$Ts = 12\pi^2 K T^2 = K \cdot 12H^2 = KR. \quad \checkmark \quad (10)$$

The key feature of this framework is that K is a thermodynamic variable, not a fixed parameter [6, 7]. This is a consequence of the Kronecker anomaly [8]: in the fully homogeneous de Sitter state, the gravitational coupling emerges as an additional zero-momentum degree of freedom.

Two remarks are in order. First, the Gibbs–Duhem relation (3) applies to the gravitational component of the two-fluid system. When matter is added, one might expect chemical-potential terms ($\mu_m n_m$) to appear. We assume that matter enters only through the shifts of the existing thermodynamic variables (K, R, T, s), not through new conjugate pairs. This is justified if the matter is treated as an external perturbation to the de Sitter vacuum rather than as a new thermodynamic species in equilibrium with it—appropriate for $\rho_m \ll \varepsilon_{\text{vac}}$. Second, the ansatz $\delta\varepsilon_{\text{vac}} = -\alpha \rho_m$ is linear in ρ_m . Higher-order terms $\beta\rho_m^2 + \dots$ are not excluded; the result $\alpha = 1/4$ applies strictly at linear order.

3 Adding matter: self-consistent perturbation

3.1 Setup

We consider a local perturbation of the de Sitter state by non-relativistic matter (dust) with density $\rho_m \ll \varepsilon_{\text{vac}}$. Under this perturbation, three quantities may shift: the vacuum energy, $\varepsilon_{\text{vac}} \rightarrow \varepsilon_{\text{vac}} - \alpha \rho_m$; the gravitational coupling, $K \rightarrow K + \delta K$; and the scalar curvature, $R \rightarrow R + \delta R$. The temperature may also shift: $T \rightarrow T + \delta T$.

The trace of the Einstein equations gives the curvature shift:

$$\delta R = 8\pi G(\rho_m + 4\delta\varepsilon_{\text{vac}}) = 8\pi G(1 - 4\alpha)\rho_m. \quad (11)$$

3.2 General case: simultaneous δT and δK

We first analyze the most general perturbation, with no variables held fixed. Writing $\delta T = \gamma \rho_m$ where γ is to be determined, and using the thermodynamic relations (8)–(9):

$$\delta\varepsilon_{\text{vac}} = 6\pi^2(2KT\gamma + T^2 \delta K/\rho_m)\rho_m = -\alpha \rho_m, \quad (12)$$

$$\delta s = 12\pi^2(K\gamma + T \delta K/\rho_m)\rho_m. \quad (13)$$

From (12):

$$\delta K = -\frac{(\alpha + 12\pi^2 KT\gamma)}{6\pi^2 T^2} \rho_m. \quad (14)$$

Substituting into (13) and simplifying:

$$\delta s = -\left(\frac{2\alpha}{T} + 12\pi^2 K\gamma\right)\rho_m. \quad (15)$$

The perturbed Gibbs–Duhem relation, to first order, requires:

$$T \delta s + s \delta T = K \delta R + \delta K \cdot R_0. \quad (16)$$

The left-hand side evaluates to (using $s = 12\pi^2 KT$):

$$\text{LHS} = T\left[-\frac{2\alpha}{T} - 12\pi^2 K\gamma\right]\rho_m + 12\pi^2 KT\gamma \rho_m = -2\alpha \rho_m. \quad (17)$$

Note: the γ -dependent terms cancel identically on the left-hand side.

The right-hand side evaluates to:

$$\text{RHS} = \frac{(1 - 4\alpha)}{2} \rho_m - 2(\alpha + 12\pi^2 K T \gamma) \rho_m. \quad (18)$$

Equating:

$$-2\alpha = \frac{1 - 4\alpha}{2} - 2\alpha - 24\pi^2 K T \gamma, \quad (19)$$

which simplifies to:

$$24\pi^2 K T \gamma = \frac{1 - 4\alpha}{2}. \quad (20)$$

This is a single equation in two unknowns (α and γ). The Gibbs–Duhem relation therefore admits a one-parameter family of solutions:

$$\gamma = \frac{(1 - 4\alpha) G}{3\pi T}, \quad (21)$$

where we used $K = 1/16\pi G$.

3.3 Physical selection: $\delta T = 0$ for local perturbations

The temperature $T = H/\pi$ is a property of the cosmological background—it is determined by the global Hubble expansion rate. A local matter perturbation (a galaxy, a cluster, a bound structure) does not modify the global H ; it modifies the local curvature R and the local vacuum energy, but the background expansion rate is set by the homogeneous cosmological solution. This is the standard separation of scales in cosmological perturbation theory: sub-horizon perturbations do not alter the background evolution.

The condition $\delta T = 0$ (i.e., $\gamma = 0$) is therefore not an arbitrary postulate but a consequence of the physical distinction between local and cosmological variables. Setting $\gamma = 0$ in Eq. (20) yields:

$$\boxed{0 = \frac{1 - 4\alpha}{2} \implies \alpha_{\text{th}} = \frac{1}{4}.} \quad (22)$$

The corresponding shifts are:

$$\delta K = -\frac{\alpha \rho_m}{6\pi^2 T^2} = -\frac{\rho_m}{24\pi^2 T^2}, \quad (23)$$

$$\delta s = -\frac{2\alpha \rho_m}{T} = -\frac{\rho_m}{2T}. \quad (24)$$

3.4 Role of the fixed- T condition

It is important to be explicit about how much work the $\delta T = 0$ condition does. In the general family (21), α ranges continuously: any α is thermodynamically admissible, provided T shifts by the corresponding γ . The condition $\gamma = 0$ is what selects $\alpha = 1/4$. Therefore, the “uniqueness” of α_{th} is not purely algebraic—it rests on the physical argument of Section 3.3.

Within the fixed- T framework, the following subsidiary results hold: the condition $\delta K = 0$ (fixed G) is inconsistent with $\rho_m \neq 0$ (the Gibbs–Duhem relation reduces to $0 = \rho_m/2$),

demonstrating that within this framework, the vacuum–matter coupling requires a local variation of K . Similarly, the condition $\delta K = 0$ with δT free gives $0 = 1$, independent of α (see Appendix).

4 Connection to QCD

4.1 The chiral fraction

The thermodynamic derivation gives $\alpha_{\text{th}} = 1/4$ as the vacuum response at linear order, under the assumptions stated above. The QCD chiral sector contributes

$$\alpha_{\chi} = \frac{\sigma_{\pi N}}{m_N} \approx 0.05. \quad (25)$$

The ratio

$$\frac{\alpha_{\chi}}{\alpha_{\text{th}}} = \frac{0.05}{0.25} = 0.20 \quad (26)$$

invites an interpretation as the chiral fraction of the vacuum response.

However, a conceptual distinction should be noted. The thermodynamic α_{th} describes the total vacuum energy response to any dust perturbation, while α_{χ} describes specifically the chiral condensate’s response to baryonic matter. These are the same quantity only if the vacuum response to matter is proportional to each sector’s contribution to the vacuum energy. If the response functions are non-proportional to the energy contributions, the ratio (26) does not directly measure the energy fraction. We present the ratio as suggestive rather than definitive.

4.2 Prediction for the gluonic sector

If the proportionality assumption holds, the remaining vacuum response is carried by the gluon condensate [10]:

$$\alpha_{\text{glue}} = \alpha_{\text{th}} - \alpha_{\chi} \approx 0.20. \quad (27)$$

Musakhanov [17] has computed the gluon-sector contribution to dark energy using the instanton liquid model in FLRW spacetime, obtaining the vacuum energy difference via the conformal anomaly. His framework provides the natural setting for computing α_{glue} : the in-medium shift of the instanton vacuum at finite baryon density would directly test Eq. (27). Independent QCD-based approaches to dark energy—from topological sectors with nontrivial holonomy [18] and from SU(3) confinement geometry [19]—support the broader principle that the observed cosmological constant is an emergent quantity determined by QCD nonperturbative physics rather than by unsuppressed vacuum fluctuations. The density dependence of QCD condensates at finite baryon density, established by Cohen, Furnstahl and Griegel [20], underlies the microscopic derivation of α_{χ} . The in-hadron condensate framework of Brodsky and Shrock [15], further developed in Ref. [16], provides the light-front foundation: QCD condensates are localized within hadrons and do not contribute to the cosmological constant globally, consistent with our picture in which the condensate shift gravitates locally.

4.3 Effective gravitational coupling and observational constraints

The local shift of the gravitational coupling is, from Eq. (23):

$$\frac{\delta K}{K} = -\frac{16\pi G \alpha \rho_m}{6\pi^2 T^2}. \quad (28)$$

In the present universe, $\rho_m/\varepsilon_{\text{vac}} \sim 0.3/0.7$, so the relative shift is $|\delta K/K| \sim \alpha \cdot \rho_m/\varepsilon_{\text{vac}} \sim 0.1$ for $\alpha \sim 0.05$.

This magnitude requires scrutiny. Lunar Laser Ranging constrains $|\dot{G}/G| < 10^{-13} \text{ yr}^{-1}$ [12], and Big Bang Nucleosynthesis constrains the value of G at $z \sim 10^9$ to within $\sim 10\%$ of its present value [13]. The shift computed here is *spatial*, not temporal: $\delta K/K$ depends on local ρ_m , not on cosmic time. The LLR and BBN constraints apply to the time derivative and the cosmological average of G , respectively, and do not directly constrain a density-dependent spatial variation at the 10% level. Spatial variation of G is constrained by post-Newtonian parameters in the solar system [14]; the relevant bound is on the Nordtvedt effect, which tests whether gravitational self-energy gravitates normally. A full analysis of these constraints in the context of density-dependent K is beyond the scope of this letter.

The modified Poisson equation in bound structures takes the form [1]:

$$\nabla^2 \Phi = 4\pi G(1 + 2\alpha) \rho_m - \Lambda_0. \quad (29)$$

This follows from substituting (1) into the Newtonian limit of the Einstein equations with a density-dependent vacuum energy: the $-\alpha\rho_m$ shift acts as an additional source term with the same spatial profile as matter, effectively enhancing G by the factor $(1 + 2\alpha)$. For $\alpha = \alpha_\chi \approx 0.05$, this gives $G_{\text{eff}} \approx 1.1 G$.

5 Relation to Volovik's modified Gibbons–Hawking entropy

In a recent paper [5], Volovik obtained a modified Gibbons–Hawking entropy for the de Sitter cosmological horizon in $(d+1)$ -dimensional spacetime: $S_H(d) = (d-1)A/8G$, which reduces to the standard $A/4G$ only for $d = 3$. This modification relies on the local temperature $T = H/\pi$ being dimension-independent.

Our result shares a common origin: the thermodynamic consistency of the de Sitter state under perturbation. In Volovik's case, the perturbation is the change of spatial dimension; in our case, it is the addition of matter. Both probe the response of the Gibbs–Duhem relation and both lead to departures from naive expectations. Extension of our calculation to general d would yield a dimension-dependent $\alpha(d)$, which we leave for future work.

6 Discussion

The result $\alpha_{\text{th}} = 1/4$ follows from two ingredients: Volovik's Gibbs–Duhem relation $Ts = KR$ for the de Sitter gravitational component, and the physical condition that local matter perturbations do not shift the cosmological temperature $T = H/\pi$. The first ingredient

is a property of Volovik’s thermodynamic framework; the second is a consequence of the separation between local and cosmological scales. Neither ingredient invokes QCD or any specific microscopic model. The result is analogous to the Clausius–Clapeyron equation: it constrains the response of the system without specifying the molecular mechanism.

The general analysis of Section 3.2 shows that the Gibbs–Duhem relation alone does not uniquely fix α ; it admits the family $\gamma = G(1 - 4\alpha)/(3\pi T)$. The physical selection $\gamma = 0$ is crucial and should be viewed as a separate assumption alongside the Gibbs–Duhem relation itself. If T acquires a matter-dependent correction (for instance, in the interior of a collapsing region where local H differs from the background), α_{th} would be modified accordingly.

Several further limitations should be noted. The matter perturbation is treated to linear order; for $\rho_m \sim \varepsilon_{\text{vac}}$, nonlinear corrections may be significant. The Gibbs–Duhem relation is an equilibrium condition; far from equilibrium it may not apply. The derivation is performed in $3 + 1$ dimensions.

7 Conclusion

Volovik’s two-fluid thermodynamics of the de Sitter state, when perturbed by local matter at linear order, determines the vacuum–matter coupling $\alpha_{\text{th}} = 1/4$ under the condition that the de Sitter temperature is set by the cosmological background. This provides a thermodynamic foundation for the ansatz $\rho_{\text{vac}} = \Lambda_0 - \alpha\rho_m$. Within this framework, the vacuum–matter coupling requires a simultaneous local variation of the gravitational coupling $K = 1/16\pi G$; the two cannot be separated.

The QCD chiral sector contributes $\alpha_\chi \approx 0.05$, which is 20% of the thermodynamic value. If the vacuum response is proportional to each sector’s energy contribution, the remaining $\sim 80\%$ resides in the gluonic sector—a prediction testable through lattice QCD calculations of the in-medium gluon condensate.

A Impossibility of $\delta K = 0$ with variable T

Setting $\delta K = 0$ in the general perturbation: $\delta\varepsilon_{\text{vac}} = 12\pi^2 K T \delta T = -\alpha\rho_m$, so $\delta T = -\alpha\rho_m/(12\pi^2 K T)$. Then $\delta s = 12\pi^2 K \delta T = -\alpha\rho_m/T$ and $s \delta T = 12\pi^2 K T \cdot \delta T = -\alpha\rho_m$.

The perturbed Gibbs–Duhem: $T\delta s + s \delta T = K \delta R$ gives $-\alpha\rho_m - \alpha\rho_m = (1 - 4\alpha)\rho_m/2$, i.e., $-2\alpha = (1 - 4\alpha)/2$, hence $-4\alpha = 1 - 4\alpha$, yielding $0 = 1$ independently of α .

This confirms that K must participate in the vacuum’s response to matter.

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